

SUBJECT- MATH CLASS-8

CHAPTER : A SQUARE AND A CUBE

DAY-1

THE LOCKER PUZZLE

Numbers follow specific rules that allow us to predict outcomes in complex situations.

- **The Puzzle:** Queen Ratnamanjuri left 100 lockers and a challenge: if you are the first to solve which remain open, you keep the fortune.
- **The Rules:** 100 people take turns; Person 1 opens all, Person 2 toggles every 2nd, Person 3 toggles every 3rd, and so on.
- **The Logic:** A locker stays open only if it is toggled an odd number of times.
- **The Secret:** Only numbers with an odd number of factors (Square Numbers) stay open.

Activity List the factors for numbers 1 to 10 and circle those with an odd number of factors.

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DAY-2

PERFECT SQUARES

A square number is the product of a natural number multiplied by itself.

- **Notation:** For any number n , we write $n \times n = n^2$.
- **Visualizing:** The area of a square is the product of its sides.
- **Perfect Squares:** 1, 4, 9, 16, 25... are called perfect squares.

Table of Squares Complete the squares for the first 30 natural numbers.

DAY-3

PATTERNS IN UNITS DIGITS

We can identify if a number is *not* a square by looking at its last digit.

- **Possible Ends:** All perfect squares end with 0, 1, 4, 5, 6, or 9.
- **Impossible Ends:** No perfect square ends with 2, 3, 7, or 8.
- **Digit Pairs:** If a number ends in 1 or 9, its square ends in 1. If it ends in 4 or 6, its square ends in 6.

Activity Write 5 numbers that are definitely not squares based on their units digit.

DAY-4

SQUARES AND ZEROS

Zeros in square numbers follow a strict parity rule.

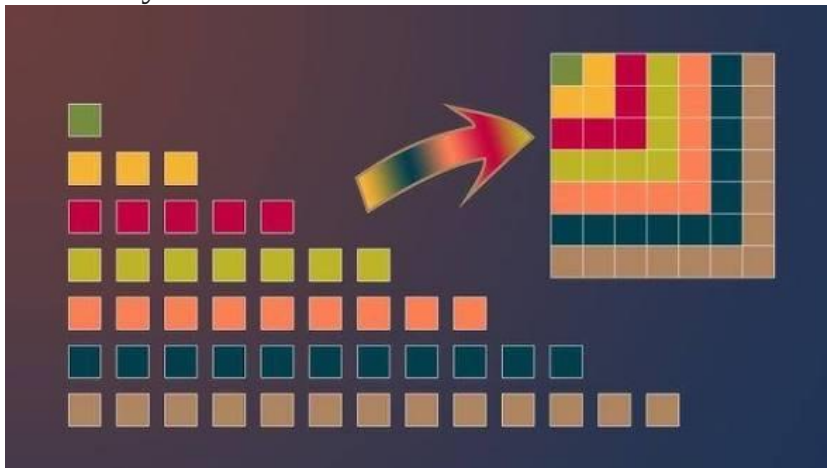
- **Doubling Zeros:** If a number has 1 zero at the end, its square has 2.
- **Even Rule:** Square numbers can only have an even number of zeros at the end.
- **Non-Squares:** A number like 2800 (with 2 zeros) must be checked further, but 250 is definitely not a square.

DAY-5

SQUARES AND ODD NUMBERS

Every square is the sum of successive odd numbers starting from 1.

- **The Pattern:** $1 = 1^2$, $1 + 3 = 4 = 2^2$, $1 + 3 + 5 = 9 = 3^2$.
- **Finding Squares:** To find 36^2 when $35^2 = 1225$, add the 36th odd number (71) to get 1296.
- **Subtracting Method:** A number is a square if successively subtracting 1, 3, 5... eventually results in 0.

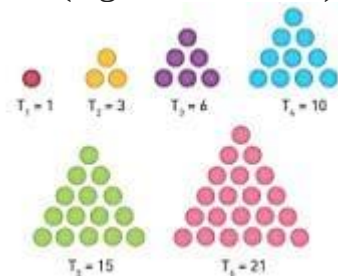


DAY-6

TRIANGULAR NUMBERS & SQUARE ROOTS

Square numbers share a deep relationship with triangular shapes.

- **Triangular Numbers:** 1, 3, 6, 10, 15....
- **Square Link:** Adding two consecutive triangular numbers gives a square number (e.g., $3 + 6 = 9$).
- **Inverse Operation:** Square root $\sqrt{x}$ is the inverse of squaring.
- **Two Roots:** Every perfect square has a positive and a negative integer square root (e.g., $\sqrt{64} = \pm 8$).



Page Number-131 – Figure it out

DAY-7

LONG DIVISION METHOD

For very large numbers where prime factorization is tedious, we use the long division method to find square roots.

- **Step 1:** Group digits in pairs starting from the units place.
- **Step 2:** Find the largest square less than or equal to the first pair.
- **Step 3:** Double the divisor and find the next digit to continue the division.

Activity Find the square root of 441 using the long division method.

DAY-8

PRIME FACTORIZATION & ESTIMATION

Finding roots through factors and educated guesses.

- **Factor Grouping:** A number is a perfect square if its prime factors can be split into two identical groups.
- **Example:** $324 = (2 \times 3 \times 3) \times (2 \times 3 \times 3) = 18^2$.
- **Estimation:** To find $\sqrt{1936}$, note it is between $40^2 = 1600$ and $50^2 = 2500$. Since it ends in 6, test 44 or 46.

Activity Find if 1156 is a perfect square using prime factorization.

DAY-9

PERFECT CUBES

A cube is a solid figure with all equal sides; its volume is the product of its side three times.

- **Definition:** A number obtained by multiplying a number by itself three times (e.g., $2 \times 2 \times 2 = 8$).
- **Perfect Cubes:** 1, 8, 27, 64, 125....
- **Prime Factors:** In a perfect cube, each prime factor appears three times or in triplets.

$$1 = 1 = 1^3$$

$$3 + 5 = 8 = 2^3$$

$$7 + 9 + 11 = 27 = 3^3$$

$$13 + 15 + 17 + 19 = 64 = 4^3$$

$$21 + 23 + 25 + 27 + 29 = 125 = 5^3$$

$$31 + 33 + 35 + 37 + 39 + 41 = 216 = 6^3$$

DAY-10

TAXICAB NUMBERS & HISTORY

Mathematics is filled with history and fascinating "dull" numbers.

- **1729:** Known as the Hardy-Ramanujan Number.
- **The Property:** It is the smallest number expressible as the sum of two cubes in two different ways ($1^3 + 12^3$) and ($9^3 + 10^3$).
- **Ancient Terms:** "Varga" was used for square and "Ghana" for cube in ancient Sanskrit. "Mula" (root of a plant) became the term for mathematical roots.

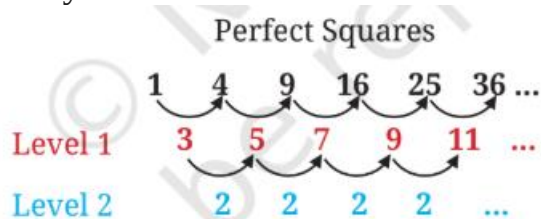
Activity Identify the two ways to express 4104 as a sum of two cubes.

DAY-11

CUBE ROOTS & SUCCESSIVE DIFFERENCES

The inverse of cubing is finding the cube root $\sqrt[3]{x}$

- **Cube Root:** If $y = x^3$, then $x = \sqrt[3]{y}$ (e.g., $\sqrt[3]{27} = 3$).
- **Pattern Levels:** For squares, differences become constant at Level 2 (always 2).
- **Cube Differences:** Compute differences for cubes (1, 8, 27, 64...) across levels until they become constant.



DAY-12

PUZZLES & RECAP

Reviewing properties and solving the "Square Pairs" challenge.

- **Summary:** Squares end in 0, 1, 4, 5, 6, 9; Cubes can end in any digit.
- **Final Challenge:** Arrange numbers 1 to 17 so every adjacent pair sums to a square (e.g., $15 + 1 = 16$).

Assessment

1. Guess the cube root of 1331 without factorization.
2. State True/False: The cube of any odd number is even.

PAGE 10

CW – Q1 – I, III, Q2, Q3 – I, III, V, Q4, Q5, Q6, Q7 – I

HW – Q1 – II, IV, Q2 – II, IV, Q8, Q9

PAGE 16

CW – Q1, Q2, Q4, Q5 – I, III

HW – Q3, Q5 – II, IV